

LLM Judge API Budget — MAM-AI v0.2 HealthBench

Two budget items for the HealthBench rubric track of the v0.2 evaluation, both via the closed-source Batch API. Starting model: `gpt-5-mini` — the smallest OpenAI option, used as a low-cost feasibility check before committing to larger spend. (Separate smaller asks of \$30 each cover the SAQ rescore and the faithfulness judge.)

Phase	Purpose	Estimated batch cost	Ask
A. Calibration	Validate <code>gpt-5-mini</code> as LLM judge against physician ground truth	\$6	\$15
B. Production rescore	Score 38,308 model responses on the HealthBench rubric track	\$33	\$50

Current ask total: \$65 (`gpt-5-mini` batch). A further ~\$50 covers 2 anticipated retrieval-module iterations (see Forward-looking section) — filed separately when each triggers, not part of this approval. Full project envelope at `gpt-5-mini`: ~\$115.

Strategy: `gpt-5-mini` is the cheapest plausible OpenAI judge. Phase A produces a metric report directly comparable to the open-weight bake-off (Background) on the same 6,853 physician-labeled triples. If `gpt-5-mini` materially beats the open-weight not-met agreement (67.9%), we run Phase B at `gpt-5-mini` for the full v0.2 report. If not, we escalate to a higher-tier closed-source judge (`gpt-5.4`, `gpt-5.5`, or alternatives — see “Escalation options” below). Either way, Phase A is the empirical gate.

Open-weight judges over-rate “met” on a meaningful fraction of physician-not-met rows (Background), biasing the deployment-headline rubric metric in the wrong direction — so a closed-source judge is necessary for the final v0.2 evaluation. The question Phase A answers is which closed-source tier suffices.

Background

MAM-AI v0.2’s open-ended evaluation tracks use an LLM judge to score Gemma 4 E4B responses against physician-written rubric criteria. We already ran an open-source bake-off across three candidates (Llama 4 Maverick FP8, Nemotron-Ultra-253B FP8-quantized, `gpt-oss-120b`) on a 6,853-row physician-labeled meta-evaluation set (`mamabench@v0.2.1/calibration/obgyn_meta_eval.jsonl`). Result: all three open candidates systematically over-rate “met” on the rows where physicians unanimously agreed the criterion was NOT met.

Judge (open-weight)	Footprint	Judge met-rate vs phys 77%	Not-met agreement	Over-rated rate on not-met rows
Llama 4 Maverick FP8	4× H200	88.8% (+11.8 pp)	44.6%	55.4%
Nemotron-Ultra-253B (FP8 q.)	4× H200	83.3% (+6.3 pp)	55.3%	44.7%
<code>gpt-oss-120b</code> (medium, pinned)	1× H200	73.2% (−3.9 pp)	67.9%	32.1%

The pinned candidate (gpt-oss-120b medium) is the best of the three. We also confirmed higher reasoning effort doesn't fix it: gpt-oss-120b at `reasoning_effort=high` improved not-met agreement by only 1.7 pp (67.9% → 69.6%) while degrading other metrics — the task is not reasoning-bound. Bigger ≠ better either: the two heavier candidates score worse, not better.

A judge that over-rates inflates the rubric headline metric (`weighted_met`) — the exact direction we don't want for a deployment story. We've exhausted the obvious open-source levers (3 models, 2 reasoning settings on the best one) and the bias persists across all of them. The final evaluation requires a closed-source judge. We start with the cheapest plausible option — gpt-5-mini — and use Phase A to test whether it materially beats the open-weight baseline on the not-met-agreement metric. If yes, Phase B runs at gpt-5-mini. If no, we escalate (see "Escalation options").

Phase A — Calibration (gpt-5-mini vs physician ground truth)

Task

Run gpt-5-mini over the OBGYN-scoped HealthBench grader meta-evaluation set (`mamabench@v0.2.1/calibration/obgy`) — 6,853 physician-labeled (`prompt`, `completion`, `criterion`) triples. Compute a 4-metric report: LLM↔single-human agreement, LLM↔consensus, judge met-rate vs physician met-rate, and per-class agreement (rubber-stamp detector).

Why necessary

The Background section establishes that open-weight judges over-rate "met" on physician-not-met rows. Before committing the Phase B spend, Phase A validates that gpt-5-mini materially improves on this metric — by running it over the same 6,853 physician-labeled triples and producing a side-by-side report against the open-weight bake-off. If gpt-5-mini doesn't pass (i.e., not-met agreement isn't materially above 67.9%), the result still has value: it tells us closed-source mini-tier is insufficient and we'd escalate to a higher tier. Phase A is the empirical gate that decides which model goes into Phase B.

Cost calculation

Pricing: gpt-5-mini Batch = \$0.125 / 1M input · \$1.00 / 1M output (50% off the standard \$0.25 / \$2.00).

Token volume (measured from actual calibration data):

	per call	× 6,853 calls
Input (grader template + conversation + completion + criterion)	~1,361	9.33 M
Output at <code>reasoning_effort=medium</code>	~700	4.80 M

Component	Batch cost
Input: 9.33 M × \$0.125/M	\$1.17
Output: 4.80 M × \$1.00/M	\$4.80
Estimated total	\$6
Ask (rounded up for safety margin on a small dollar amount)	\$15

Phase B — Production Rescore (HealthBench rubric, ±RAG)

Task

For each (model response, criterion) pair across the 3 HealthBench configs and both ±RAG arms, ask the pinned judge whether the criterion is met. The HealthBench `weighted_met` headline metric is computed per response from these binary verdicts.

Why necessary

This is the published HealthBench evaluation methodology. Without it, the v0.2 report cannot: - Quantify Gemma’s clinical-rubric quality (the deployment-relevant metric). - Compare ±RAG on the rubric track (key for the RAG deployment claim). - Compare against the published HealthBench leaderboard.

A closed-source judge is required because the open-weight bake-off (see Background) showed every open candidate biases the headline metric upward by over-rating “met.” The judge tier validated in Phase A is the one that goes into production here. Starting tier is `gpt-5-mini`; the cost calc below uses `gpt-5-mini` batch pricing. If Phase A flags `gpt-5-mini` as insufficient, Phase B’s cost shifts to the chosen escalation tier (see “Escalation options” for tier-specific Phase B totals).

Cost calculation

Call count (one judge call per criterion, matching HealthBench’s published methodology):

Config	rows × avg criteria/row	criterion calls / arm
healthbench_oss_eval	1,209 × 11.7	14,195
healthbench_consensus	872 × 2.2	1,909
healthbench_hard	258 × 11.8	3,050
Per arm		19,154
Total × 2 arms (±RAG)		38,308

Token volume:

	per call	× 38,308 calls
Input	~1,361	52.14 M
Output at <code>reasoning_effort=medium</code>	~700	26.82 M

Component	Batch cost (<code>gpt-5-mini</code>)
Input: 52.14 M × \$0.125/M	\$6.52
Output: 26.82 M × \$1.00/M	\$26.82
Estimated total	\$33
Ask (with ~50% buffer for output-token variance)	\$50

Forward-looking estimate — future retrieval-iteration runs

The asks above cover one full v0.2 evaluation of the current MAM-AI app. Over the coming months, planned retrieval-module improvements will trigger re-runs of the +RAG arm only — the no-RAG baseline doesn’t change because it doesn’t use retrieval.

Each retrieval improvement re-runs roughly half of Phase B (one arm of 19,154 criterion calls instead of both arms' 38,308), plus a small SAQ re-run. Cost at the starting `gpt-5-mini` tier:

Per improvement	Calls	Batch cost
HealthBench rubric, +RAG arm only	19,154	~\$17
SAQ, +RAG arm only	369	~\$1
Per-improvement total		~\$18

Anticipating 2 retrieval-module improvements over the coming months:

	Cost
Per improvement	~\$18
× 2 improvements	~\$36
Forward-looking estimate (with buffer)	~\$50

This is not part of the current ask — each improvement would be a separately approved follow-up. Visibility into the full retrieval-iteration roadmap at `gpt-5-mini`: ~\$65 for the current v0.2 evaluation, then ~\$25 per retrieval improvement. (If we escalate to a higher tier after Phase A, these forward-looking numbers scale up proportionally — see “Escalation options” for per-tier project totals.)

Escalation options if `gpt-5-mini` calibration falls short

If Phase A shows `gpt-5-mini` doesn't materially beat the open-weight baseline (67.9% not-met agreement), we escalate to a higher-tier closed-source judge. The table below is the reference for the new ask we'd file in that case — Phase A is re-run at the escalation tier (~\$15–\$120 depending on tier), and Phase B runs at the same tier.

Token volumes are fixed regardless of vendor or tier (the workload is the same); only price-per-token differs:

Phase	Calls	Input tokens	Output tokens (at <code>reasoning_effort=medium</code>)
A — Calibration (one-time)	6,853	9.33 M	4.80 M
B — Production rescore (one-time, both $\pm$ RAG arms)	38,308	52.14 M	26.82 M
Per future retrieval iteration (+RAG arm only, HealthBench + SAQ)	19,523	27.55 M	13.78 M

Per-candidate cost — pricing + project total (1× Phase A + 1× Phase B + 2× future iterations)

Grouped by vendor (OpenAI → Anthropic → Google). Prices per 1 M tokens. All three vendors offer a Batch API at 50% off standard — the prices shown are those batch rates.

Vendor / Tier	Input \$/M	Output \$/M	Project total	+~12% buffer
OpenAI gpt-5-mini (starting point)	\$0.125	\$1.00	\$74	~\$85
OpenAI gpt-5.4-mini	\$0.375	\$2.25	\$177	~\$200
OpenAI gpt-5	\$0.625	\$5.00	\$369	~\$420
OpenAI gpt-5.4	\$1.25	\$7.50	\$590	~\$660
OpenAI gpt-5.5	\$2.50	\$15.00	\$1,181	~\$1,320
Claude Opus 4.7	\$2.50	\$12.50	\$1,031	~\$1,160
Claude Sonnet 4.6	\$1.50	\$7.50	\$618	~\$700
Gemini 3.1 Pro Preview	\$1.00	\$6.00	\$471	~\$530

Escalation recommendation

If gpt-5-mini's Phase A result is insufficient, the natural escalation order is:

1. OpenAI gpt-5.4-mini (~\$200 full project) — next step up in the mini tier; double the cost, materially more capable. Best first escalation if mini failed by a small margin.
2. OpenAI gpt-5 (~\$420 full project) — older base flagship, still recognizable as a “frontier OpenAI” name. Best if 5.4-mini is also insufficient and you want to stay in the OpenAI lineage.
3. OpenAI gpt-5.5 (~\$1,320) or Claude Opus 4.7 (~\$1,160) — frontier-tier; pick if all the mid-tier options fail the not-met threshold.
4. Gemini 3.1 Pro Preview as a \$0-out-of-pocket alternative — frontier-class Google model; new Google Cloud accounts get \$300 in free credits each on sign-up, so stacking 2 fresh accounts covers the ~\$530 project cost entirely. Tradeoff is logistical (multiple new accounts with separate credit cards), not financial.

In all cases, each escalation would be a separate ask filed after Phase A produces the data justifying it.